

Solver Documentation
electroChem
simplePhi branch

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1 Introduction

This document documents the custom OpenFOAM solver electroChem made for simulating reacting multicomponent species electrolytic systems with the goal of supporting microbial electrosynthesis of CO₂ to acetate. The solver is an extension of the existing pimpleFOAM FVM solver from OpenFOAM.

2 simplePhi branch

This branch implements resolved potential based on a constant applied current density. This allows for an algebraic solution of the potential gradient in the electrode normal direction rather than requiring the full Poisson equation for potential. concentration and potential are coupled values but for this model they are treated as a staggered explicit solution, assuming the time-step is sufficiently small to satisfy stability. Later this assumption can be changed for an iterative correction loop.

3 Governings equations

The flow is treated as incompressible and solved through the incompressible Navier Stokes formulation using the PIMPLE algorithm

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) = -\frac{1}{\rho} \nabla P + \nu \nabla^2 \mathbf{u} + \mathbf{g} \quad (2)$$

All active species are treated by reactive Nerst-Planck

$$\frac{\partial c_i}{\partial t} + \nabla \cdot (c_i \mathbf{u}) = \nabla \cdot (D_i \nabla c_i) + \nabla \cdot \left(D_i \frac{z_i F}{RT} c_i \nabla \phi \right) + R_i \quad (3)$$

Where R_i is the reactive species source term combining the contributions of all reactions

$$R_i = \sum_j \nu_{ij} \dot{r}_j \quad (4)$$

where ν_{ij} is the stoichiometric constant of species "i" for reaction "j". A constant current density is prescribed over the domain in the electrode normal direction $\mathbf{e}_J$

$$\mathbf{j} = j_{app} \cdot \mathbf{e}_J \quad (5)$$

With this the potential gradient can be algebraic updated from the current equation

$$\mathbf{j} = -F \sum_i z_i D_i \nabla c_i - \frac{F^2}{RT} \sum_i z_i^2 D_i c_i \nabla \phi \quad (6)$$

with

$$\kappa = \frac{F^2}{RT} \sum_i z_i^2 D_i c_i \quad (7)$$

and

$$\mathbf{j}_D = -F \sum_i z_i D_i \nabla c_i \quad (8)$$

resulting in

$$\nabla \phi = \frac{\mathbf{j}_D - \mathbf{j}}{\kappa} = \frac{\mathbf{j}_D - j_{app} \cdot \mathbf{e}_J}{\kappa} \quad (9)$$

Species are converted to mass fraction

$$Y_i = c_i \frac{M_i}{\rho} \quad (10)$$

and the inert species is explicitly closed by

$$Y_{inert} = 1 - \sum_{i \neq inert} Y_i \quad (11)$$

4 Custom boundary conditions

The electrode flux is comprised of a faradaic source combined with diffusive and migration flux assuming no penetration of the electrode.

$$\nabla c_i \cdot \mathbf{n} = \frac{\nu_i j_{el}}{D_i n F} - \left(\frac{z_i F}{RT} c_i \nabla \phi \right) \quad (12)$$

Discretization

The boundary is solved using a Robin equation formulation. Introduce

$$A = \frac{\nu_i j}{n F D_i}, \quad \beta = \frac{z_i F}{RT} (\nabla \phi \cdot \mathbf{n}) \quad (13)$$

At the electrode boundary

$$\nabla c_i \cdot \mathbf{n} = A - \beta c_{b,i} \quad (14)$$

where $c_{b,i}$ is the boundary concentration of species i . Then the boundary adjacent gradient is defined as

$$\nabla c_i = \delta (c_{b,i} - c_{p,i}) \quad (15)$$

where $\delta = 1/\Delta_n$, Δ_n is the distance from boundary to cell-center and $c_{p,i}$ is the boundary adjacent cell-center concentration. Insert into equation 14

$$\delta (c_{b,i} - c_{p,i}) = A - \beta c_{p,i} \quad (16)$$

isolate boundary concentration

$$c_{b,i} = \frac{\delta c_{p,i} + A}{\delta + \beta} \quad (17)$$

and then the boundary flux becomes

$$\nabla c_i \cdot \mathbf{n} = A - \beta c_{b,i} \quad (18)$$

5 Reaction Models

The solver supports kinetic reactions treated as source terms or instantaneous equilibrium projections applied after species transport.

5.1 Kinetic Reactions

Kinetic reactions are for reactions developing on timescales comparable or slower than that of the transport and must be treated explicitly.

5.1.1 Reversible kinetic reaction

The reaction rate of reversible kinetic reactions are found as

$$\dot{r} = k_f \prod c_R - k_b \prod c_P \quad (19)$$

where k_f is the forward reaction rate and k_b is the reverse reaction rate. c_R is the concentration of reactants and c_P is the concentration of products.

5.1.2 Multiplicative Monod reaction

Microbial reactions are described through multiplicative Monod

$$\dot{r} = X \dot{q}_{max} \left(\frac{c_d}{c_d + K_{s,d}} \right) \left(\frac{c_a}{c_a + K_{S,a}} \right) \quad (20)$$

where X is the biomass density, $\dot{q}_{max}$ is the maximum specific consumption rate, K_S is the half saturation constant and the subscripts d and a describe the electron donor and acceptor substrates.

5.2 Equilibrium Reactions

Equilibrium reactions assume that the equilibrium settles much faster than the transport time scale and can therefore be assumed instantaneously before the next transport step.

5.2.1 Water equilibrium

Water equilibrium treats the reactions



with equilibrium constant

$$K_W = c_{H^+} c_{OH^-} \quad (22)$$

As the water equilibrium reaction can only create or consume ion pairs the ion difference must be conserved

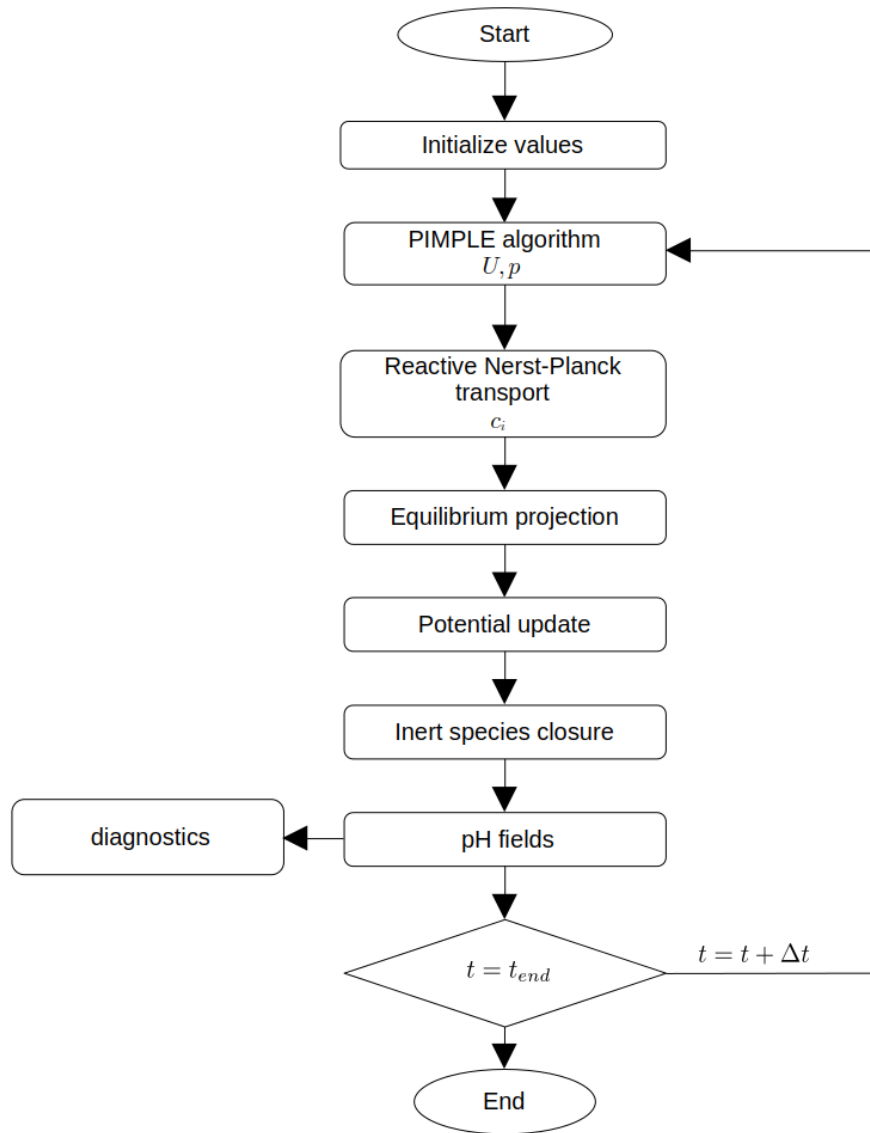
$$A = c_{H^+} - c_{OH^-} \quad (23)$$

Rewriting and substituting equation 23 into equation 22 the following algebraic concentration projections are found

$$c_{H^+} = \frac{A + \sqrt{A^2 + 4K_w}}{2} \quad (24)$$

$$c_{OH^-} = \frac{-A + \sqrt{A^2 + 4K_w}}{2} \quad (25)$$

6 Algorithm flowchart



7 Solver structure

Source code structure

```
electroChem/  
|-- Documentation/  
|-- Make/  
|   |-- files  
|   |-- options  
|-- reactionModels/  
|   |-- equilibriumReactions/  
|       |-- waterEquilibrium.C  
|       |-- waterEquilibrium.H  
|   |-- kineticReactions/  
|       |-- multiplicativeMonod.C  
|       |-- multiplicativeMonod.H  
|       |-- reversibleKineticReaction.C  
|       |-- reversibleKineticReaction.H  
|-- cEqn.H  
|-- createFields.H  
|-- diagnostics.H  
|-- electroChem.C  
|-- pEqn.H  
|-- phiEqn.H  
|-- UEqn.H  
|-- YEqn.H
```

custom functions

```
src/  
|-- customLibs/  
|   |-- electrochemistry/  
|       |-- derivedFvPatchFields/  
|           |-- MolarFlux/  
|               |-- faradaicMolarFlux.C  
|               |-- faradaicMolarFlux.H  
|               |-- robinMolarFlux.C  
|               |-- robinMolarFlux.H  
|-- Make/  
|   |-- files  
|   |-- options
```